The γ-flash time base of the n_TOF EAR2 X17 campaign

A calibration from the seven runs taken with the SiPM-wall divert disabled

n_TOF X17 / DREAM analysis

2026-07-28

Abstract

The SiPM walls of the X17 setup are blanked around the γ-flash, so they never record it, and the pulse-shape analysis (PSA) that produces the official hit files mis-times the flash on every detector except the beam pickup. We identify the seven runs of the 2026 EAR2 campaign in which the blanking gate was inadvertently disabled — the only direct view of the true flash arrival in the walls — and use them to calibrate the flash time of all 32 wall channels against the beam pickup (PKUP):

$$t_{\text{flash}}(\text{bunch}) \;=\; t_{\text{PKUP}}(\text{bunch}) + C, \qquad C \simeq -1719\,\text{ns}.$$

The constant reproduces to 0.5 ns between runs of one epoch, and the liquid scintillators — which are never gated — show that the underlying time base is stable to 0.68 ns rms over the whole campaign. We further show that the 5 ns shift between parasitic and dedicated proton pulses is an artefact of front-end saturation rather than a change of the flash arrival time, and we quantify why the plastic scintillators cannot be used for flash timing.

1 Result

The calibration is a per-channel constant C referred to the pickup pulse of the same bunch:

$$t_{\text{flash at detector}}(\text{bunch}) = t_{\text{PKUP}}(\text{bunch}) + C[\text{channel}]. \tag{1}$$

Table 1 gives the per-wall values; the per-channel numbers are in `data/flash_timing_calibration.json`. The recommended set is the 2026-07-16 epoch, for runs $\geq$ 224400.

Table 1: Wall constants $C = t_{\text{channel}} - t_{\text{PKUP}}$, in ns. "spread" is the r.m.s. over the eight channels of a wall.

wall	all 7 runs	2026-07-11	**2026-07-16**	channel spread	per-bunch σ	flash amp. [ADC]
WALA	−1717.5	−1716.7	**−1719.5**	5.2	3.2	28613
WALB	−1719.3	−1719.1	**−1719.7**	3.2	3.1	28515
WALC	−1715.5	−1714.0	**−1719.2**	1.9	3.6	28295
WALD	−1714.1	−1712.2	**−1719.0**	3.9	3.4	28246
all 32	−1716.6	−1715.5	**−1719.3**	3.4	3.5	28420

Do not use the stored `tflash`. In gated runs the PSA flash finder is bistable *within a single run*: in some bunches it tags the blanking-gate transient, in others the clamped leak at the true flash time. No per-run offset can repair that. The pickup finder, by contrast, never fails (§4.1).

Table 2: Uncertainty budget.

effect	size [ns]
run-to-run reproducibility within an epoch	0.5
time-base stability over the campaign (LIQ common mode, r.m.s.)	0.68
transport from the 07-16 epoch to the end of the campaign	1.2
07-11 → 07-16 epoch difference (real hardware change)	3.9
beam-intensity dependence, parasitic → dedicated	5.0
single-bunch spread, one channel	3.5
single-bunch spread, 32-channel average	2.5

2 Why special runs were needed

The wall signal is diverted around the flash by a $\sim$1 μs blanking gate (N1081B module 6, section C). The walls therefore record either the gate transient — 1190 ns before the flash in the July-11 configuration, 375 ns after the gate delay was changed from 200 to 1000 ns on 2026-07-22 — or a clamped leak at the true flash time.

A scan of every processed run of the campaign (306 runs, 224264–224587) identifies exactly seven runs in which the gate was inactive (Table 3): all four walls then tag the same time to within $\pm$6 ns and the flash peak reaches $\sim$28 500 ADC instead of the $\sim$850 ADC transient. The 07-11 group corresponds to the switch outage during which the N1081B module 6 was offline; the 07-16 pair coincides with the wall/plastic recalibration. 145 runs before 224345 have no wall trees in the official processing and cannot be checked without a tape reprocessing.

Table 3: The seven divert-off runs.

run	date	bunches	intensity	all-four-walls agree
224356	07-10 21:44 – 07-11 01:00	3412	dedicated	85 %
224357	07-11 01:01 – 03:51	3286	47 % ded. / 53 % par.	98 %
224358	07-11 03:52 – 06:40	3269	mixed	93 %
224359	07-11 06:41 – 09:40	3426	dedicated	87 %
224360	07-11 09:41 – 10:08	549	dedicated	91 %
224464	07-16 09:54 – 12:56	2716	mostly dedicated	60 %
224466	07-16 $\sim$13:00 – 15:19	2073	mostly dedicated	100 %

3 Method

For every (bunch, channel) we take the largest-amplitude hit within $\pm$300 ns of that bunch's flash anchor and subtract the pickup pulse time of the same bunch. The quoted constant is the mean over bunches after a $\pm$100 ns core cut that removes PSA mis-tags. Since `tof` is quantised to 1 ns, the mean over $\sim$3000 bunches has a statistical error of 0.06 ns: every number below is limited by systematics, not statistics.

4 Per-channel constants and resolution

On 07-11 the four walls sat 7 ns apart; by 07-16 they agree to 0.7 ns. The shift is per-channel (WALB $-$0.6 ns, WALD $-$6.9 ns), i.e. a front-end change rather than a cable change, consistent with the wall threshold/HV equalisation of 07-15/16. The equalised state is the one that held for the rest of the campaign, which is why the 07-16 constants are recommended.

Decomposing the 32-channel matrix bunch by bunch gives, averaged over the seven runs: 3.5 ns for a single channel against the pickup, of which 2.5 ns is common to all 32 channels and

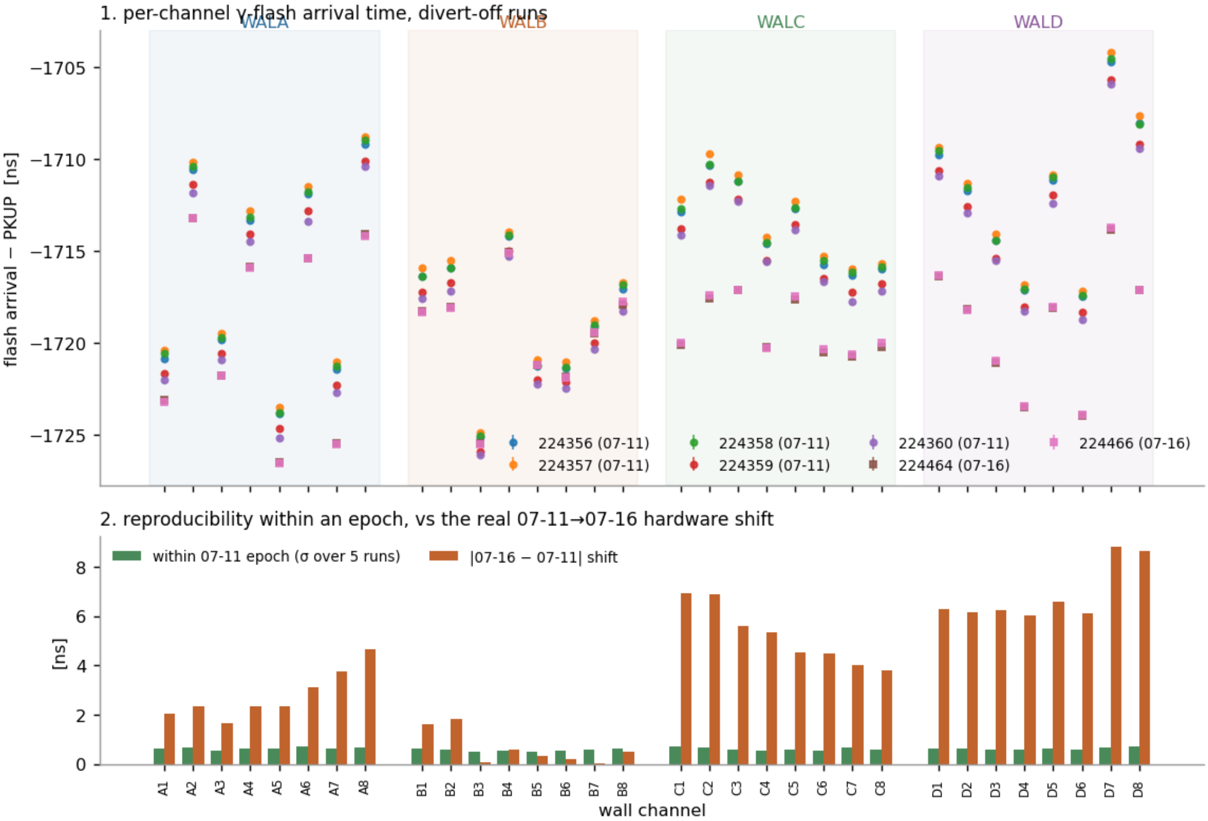

Figure 1: Per-channel flash arrival referred to the pickup, for all seven runs (top), and the decomposition of the run-to-run scatter into within-epoch reproducibility and the real 07-11→07-16 hardware shift (bottom). Within an epoch every channel reproduces to ∼0.6 ns; the epoch shift is channel-dependent and reaches 9 ns on WALD.

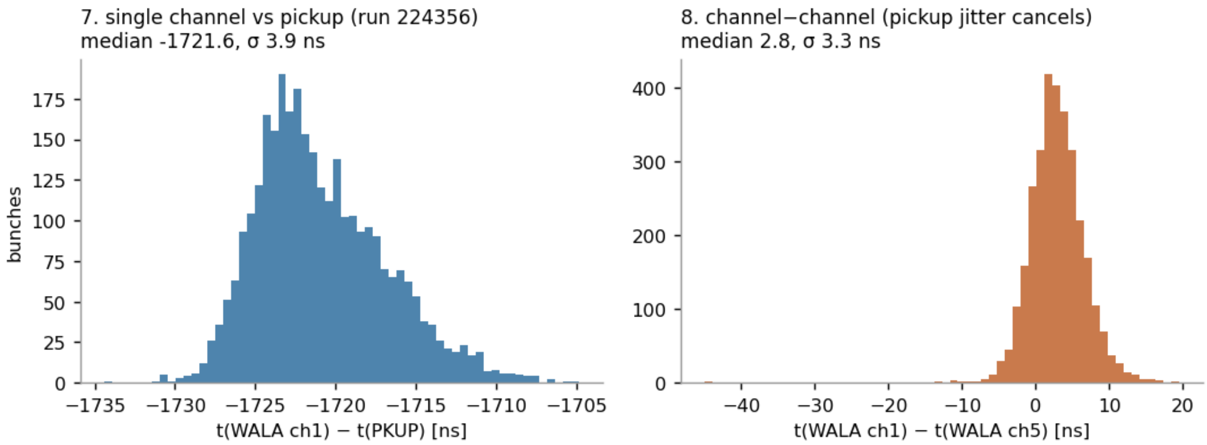

Figure 2: Left: single channel against the pickup. Right: the difference of two channels of the same wall, in which the pickup and beam jitter cancel.

2.5 ns is independent. The independent part averages away over the array; the common part does not, and since the pickup itself contributes only ∼1 ns (§4.1) it represents a genuine bunch-to-bunch variation of the flash arrival. The 07-16 runs are better than the 07-11 ones (2.7 vs 3.9 ns per channel), again following the equalisation.

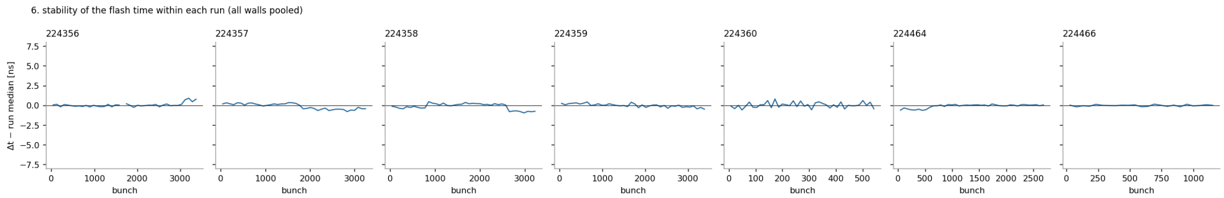

Figure 3: Within-run stability: pooling all 32 channels in 40 bins across a full three-hour run, the bin-to-bin spread is 0.25–0.39 ns and the maximum excursion 1.4 ns.

4.1 The reference: the beam pickup

The pickup is a near-ideal reference: its pulse time has a bunch-to-bunch spread of 0.83 ns (dedicated) to 1.42 ns (parasitic), it is usable in 98.4 % of bunches — the 1.6 % of failures are all parasitic pulses — its amplitude is strictly linear in protons (ratio 1.97 for a factor two), and its timing moves by only −0.45 ns between the two intensity classes.

5 Is the intensity dependence real?

The wall constant differs by −5.0 ns between parasitic (4.1×10^{12} p) and dedicated (8.5×10^{12} p) pulses. Three independent arguments show that this is a property of the measurement, not a change of the flash arrival time.

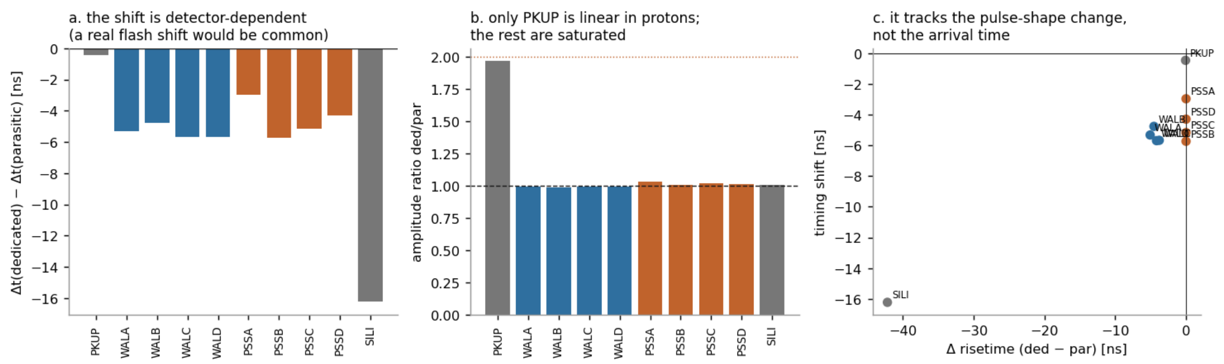

Figure 4: (a) The shift is strongly detector-dependent — 0 for the pickup, −5 ns for the walls, −3 to −6 ns for the plastics, −16 ns for the silicon monitor. A genuine change of the flash arrival would be common to all. (b) Only the pickup is linear in protons; every other detector is saturated, its amplitude ratio being ≃ 1. (c) The shift follows the change in pulse risetime.

1. **It is detector-dependent.** The same flash is seen by all detectors, so a real change in its arrival time would shift them equally. Instead the shift ranges from 0 (pickup) through −5 ns (walls) to −16 ns (silicon monitor, which is the most deeply saturated, with the PSA saturation flag set on 78 % of its flash hits).
2. **The pickup does not move.** The pickup timestamps the very protons that produce the flash, and its amplitude doubles between the two classes while its time moves by −0.45 ns. The flash cannot precede its own protons.

3. **Classical amplitude walk is excluded quantitatively.** Within a single intensity class the dependence of the hit time on its amplitude is measurable; extrapolating it to the observed change of median amplitude between classes predicts +0.3 ns for the walls and <0.4 ns for the others — with the wrong sign for the walls (Fig. 5). The measured shifts are ten to fifty times larger.

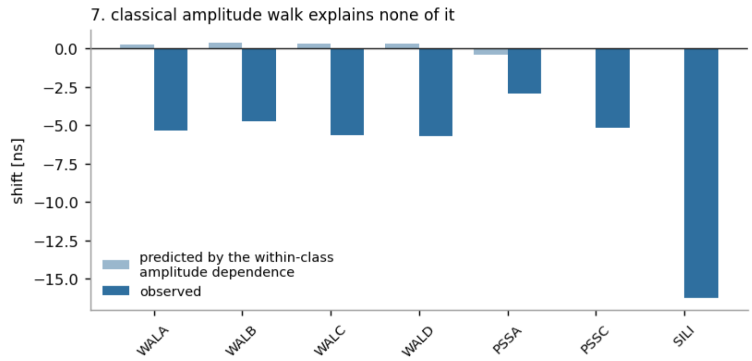

Figure 5: The amplitude dependence measured *within* one intensity class cannot account for the shift *between* classes.

The mechanism is saturation of the front end. The peak amplitude cannot grow with the flash size, so a larger flash instead reaches the clipping level earlier and stays there longer: on the walls the reported risetime falls from 20.5 to 15.4 ns (−25 %) while the FWHM grows from ∼700 to ∼2000 ns; on the silicon monitor the risetime falls from 98 to 56 ns. Any edge-based estimator therefore fires earlier. *Practical consequence:* the effect must be carried in the calibration (use the constant matching the pulse type, or apply the −5 ns per class), but it must not be interpreted as physics.

6 Stability in time

Because the walls can only be measured in the seven divert-off runs, the stability of the time base is established with the liquid scintillators and the plastics, which are never gated and therefore see the flash in every run.

Over 36–39 runs spanning 224400–224584 (2026-07-13 to 07-28) the four liquid cells give

$$\sigma_{\text{run}}(\text{LIQA}) = 0.73\,\text{ns}, \quad \sigma_{\text{run}}(\text{LIQB}) = 0.73\,\text{ns}, \quad \sigma_{\text{run}}(\text{LIQC}) = 0.77\,\text{ns}, \quad \sigma_{\text{run}}(\text{LIQD}) = 1.48\,\text{ns}.$$

The four cells move *together*: the within-run spread of their deviations is 0.22 ns and the LIQA–LIQC correlation across runs is +0.92, so what is observed is a common wander of the time base rather than detector noise. Its r.m.s. is 0.68 ns and the largest single-run excursion is −2.2 ns (run 224531). The two divert-off runs of 07-16 sit +0.9 to +1.2 ns from the late-campaign mean, which is the figure quoted in Table 2 for the transport of the wall calibration to the end of the campaign.

Independent check in gated runs. Where the gate is active the walls still emit a clamped leak. Measured against Eq. (1) in runs 224362 (07-11 era) and 224572 (07-26, fifteen days after the nearest calibration run and after the gate-delay change), the leak lands within 6.2 ns on average and 9.2 ns worst case on all 32 channels. The gate transient itself sits at −1190 ns and −375 ns respectively, reproducing the +800 ns delay change of 07-22.

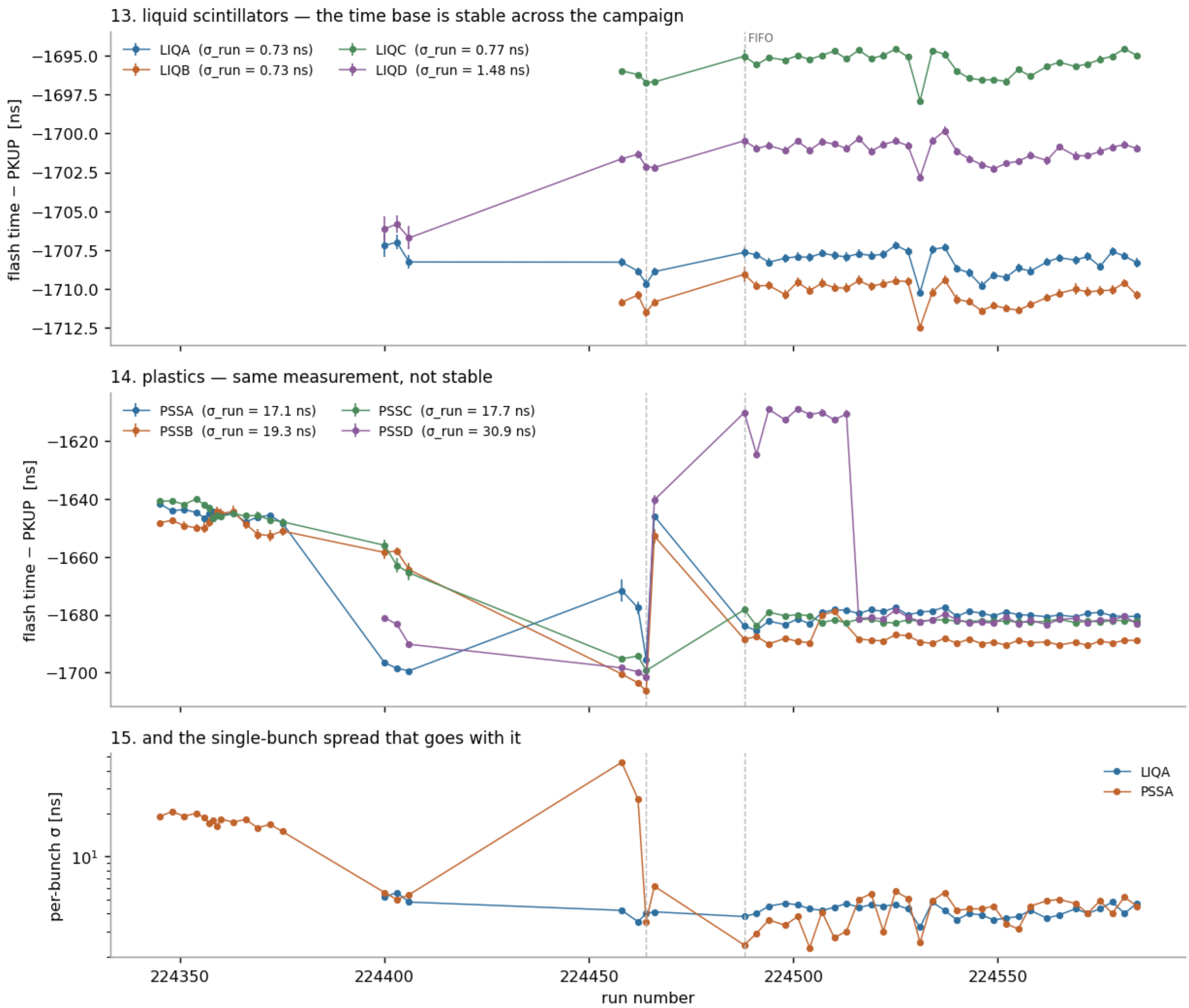

Figure 6: Flash time relative to the pickup, run by run, with statistical errors; bottom panel: the single-bunch spread. The liquids are stable, the plastics are not.

7 Why the plastics cannot be used — in the official processing

Note added. Everything in this section describes the *official* processed files. The parallel UserInput work (`FINDINGS_2026-07-28_psa_optimization.md`) reprocessed run 224572, and on the `v4_walshapes` variant the plastic flash hit is found in **100 %** of bunches with $\sigma = 3.0$–4.6 ns, against 42 % and 2.5 ns officially. The single change responsible is the **G-FLASH threshold** ($50 \to 2000/10^4$): the `v1_flash` variant, with the elimination window untouched, already reaches 100 %. The missing hits were therefore a downstream symptom of the flash-finder bug — with the flash mis-located at ∼130 ns (fitting noise), the pulse recognition for that bunch never reconstructs the real pulse at 11.63 μs. Widening the elimination window (`v2_elim`) changes the yield not at all, and the surviving flash hits have area/amp ≃ 72–76, above even the widened window — so elimination is not the gate.

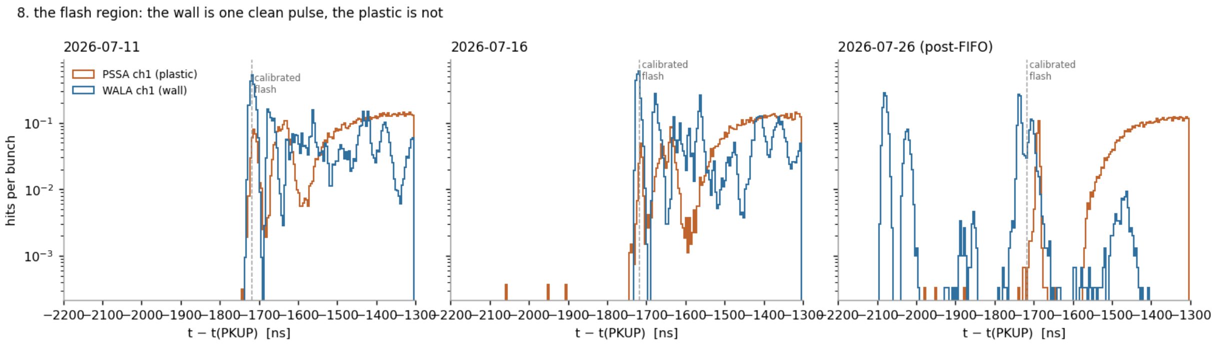

Figure 7: Hit density per bunch in the flash region. The wall produces one clean pulse; the plastic response is spread over ∼100 ns and accompanied by a continuum of fragments. In the post-FIFO run the wall shows both the gate transient (~ -2100 ns) and the leak (~ -1750 ns).

The plastics fail in three distinct ways, all quantified in `data/plastic_pathology.csv`:

Table 4: Flash-region behaviour by detector family and epoch. "found" is the fraction of bunches in which the PSA put *any* hit within 60 ns of the modal flash position; σ and the pulse quantities refer to the largest-amplitude hit there; "n_{hit}" counts hits per bunch in the $\pm 1\,\mu$s flash window.

epoch	family	found	σ [ns]	amplitude	sat. flag	FWHM [ns]	n_{hit}/bunch
2026-07-11	WAL	100 %	4.1	28521	0 %	1527	12.4
2026-07-11	PSS	**67 %**	**19.1**	23193	0 %	11	38.2
2026-07-16	WAL	100 %	2.3	28560	0 %	2602	12.5
2026-07-16	PSS	64 %	5.8	34571	0 %	43	26.0
2026-07-16	LIQ	**100 %**	4.1	**63931**	**80 %**	103	43.6
2026-07-26	WAL	100 %	8.3	19650	6 %	57	11.7
2026-07-26	PSS	**42 %**	2.5	55114	27 %	76	24.3
2026-07-26	LIQ	**100 %**	4.5	**63921**	**79 %**	102	44.1

(i) Yield. A hit at the flash time is reconstructed in only 64–67 % of bunches, falling to **42 %** after the FIFO fan-in was installed on 2026-07-17, when the plastic flash pulse begins to rail the digitiser (median amplitude 55 100, up to 63 600 of a 65 536 range). In the bunches where it is missing, the PSA reconstructs *no hit at all* before ~ -1600 ns; we verified that this is not an artefact of the extraction window, which covered the flash region in 100 % of those bunches.

The liquids are the control — and they exclude saturation as the cause. LIQ sits on the same digitiser and the same PSA, and its flash hits are *more* saturated than the plastics' (saturation flag on 79–80 % against 27 %, amplitude 63 900 against 55 100), with the highest hit multiplicity of all (44 per bunch). Yet the PSA finds the liquid flash in **100 %** of bunches at $\sigma = 4.1$–4.5 ns. The plastic drop-out is therefore specific to the plastic channel — its PSA configuration and/or its pulse shape — and not a generic consequence of a railed pulse, which suggests it can be repaired in the `UserInput`.

(ii) Amplitude-dependent bias and spread. When a hit is found, its time depends on its amplitude: in run 224357 the median moves by 17 ns between the lowest and highest amplitude bins, with a per-bunch σ of 7–15 ns (Fig. 8). After the FIFO the precision improves markedly ($\sigma = 2.4$–3.4 ns, bias 4 ns) but the yield collapses, so the two regimes trade against each other.

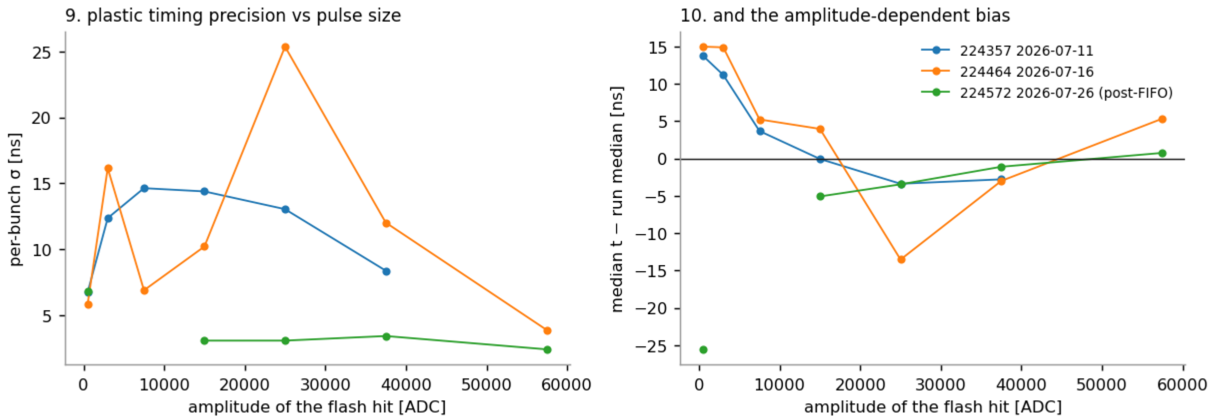

Figure 8: Plastic timing precision (left) and bias (right) versus the amplitude of the flash hit, for the three epochs.

(iii) Instability in time. The plastic constant moves by tens of nanoseconds over the campaign ($\sigma_{\text{run}} = 17$–31 ns, Fig. 6) as the plastic HV was equalised (07-16) and the FIFO fan-in installed (07-17). Channel PSSB ch2 is defective throughout (amplitude 9 778 against ∼30 000 elsewhere). Plastic constants must therefore be taken per run, never per epoch.

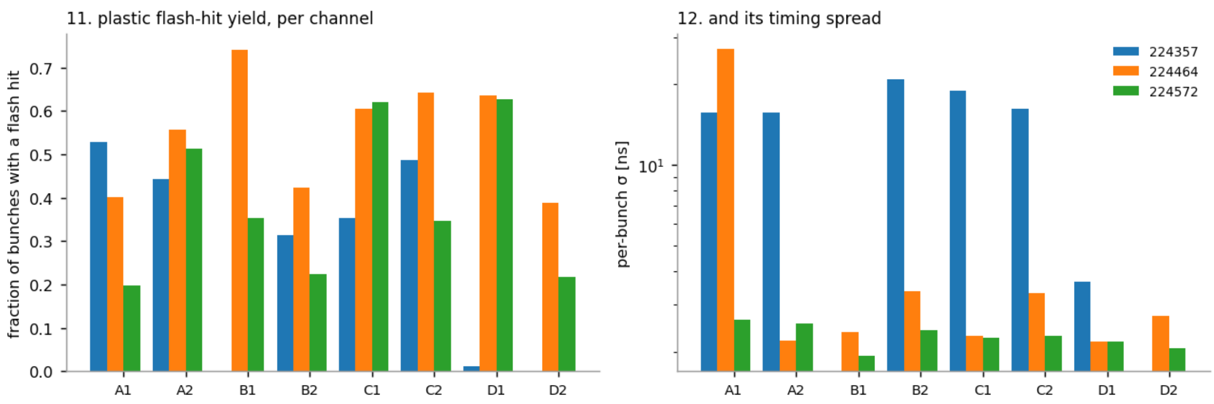

Figure 9: Per-channel flash-hit yield and timing spread for the eight plastic readouts in the three reference runs.

8 Independent cross-check against the reprocessed data

The reprocessing provides a second, independent route to the same constants: a *corrected* flash finder, run on 224572 — a gated run, ten days after the nearest calibration run, in which the walls can only see the clamped leak. Table 5 compares its stored `tflash`, referenced to the pickup (8 partials, 1597 bunches), against this work.

Table 5: Reprocessed (`v4_walshapes`) stored `tflash` minus the pickup, against the constants of this work. All values in ns.

tree	reprocessed	this work	difference	this work is from
WALA	−1717.82	−1719.46	+1.64	divert-off runs, 07-16 epoch
WALB	−1718.07	−1719.69	+1.62	divert-off runs, 07-16 epoch
WALC	−1718.05	−1719.23	+1.18	divert-off runs, 07-16 epoch
WALD	−1722.29	−1719.01	−3.28	divert-off runs, 07-16 epoch
		mean +0.29, rms	**2.07**	
LIQA	−1708.38	−1708.22	−0.16	campaign monitor (official proc.)
LIQB	−1710.78	−1710.30	−0.48	campaign monitor (official proc.)
LIQC	−1695.73	−1695.56	−0.17	campaign monitor (official proc.)
LIQD	−1701.27	−1701.56	+0.29	campaign monitor (official proc.)
		mean −0.13, rms	**0.27**	

The walls agree to 2.1 ns rms and the liquids to 0.27 ns rms — the latter being two different processings of the same run agreeing to a quarter of a nanosecond. Both sit inside the transport budget of Table 2, and the two routes share no assumption beyond the pickup itself.

9 How to apply, and what limits it

1. **Walls**: use the per-channel `C_2026_07_16` for runs $\geq$ 224400, `C_2026_07_11` before that. Add −5 ns if the pulse type differs from the one the constant was taken on.

2. **Liquids**: the per-run value, or the campaign mean; they agree to ∼1 ns.

3. **Plastics**: per-run value only.

4. Never use the stored `tflash` of WAL/PSS/LIQ *from the official files*. In reprocessed files it is sound, but PKUP + C remains the recommended time base: it is measured on the undiverted flash, whereas even a reprocessed wall `tflash` times the clamped leak of a gated signal.

5. If a bunch has no usable pickup pulse (1.6 %, all parasitic), drop the bunch rather than falling back on `tflash`.

Limitations. (i) The undiverted flash saturates the wall front end, so these runs give timing only — the energy scale must still come from the ^{88}Y runs 224476–224479. (ii) The 3.9 ns shift between 07-11 and 07-16 is real and per-channel, so wall front-end timing can move; nothing guarantees that it did not move again between 07-16 and 07-28 beyond the liquid-scintillator evidence that the shared time base did not. (iii) Runs before 224345 are unverified. (iv) The raw data of all seven runs has aged off EOS disk; a waveform-level study needs a tape stage.